



PIERO ENGINE

Time of Day

Real-World Turbidity Values

- **1.0 to 2.0 (Arctic/Mountain Clear):** The absolute minimum. The air is pristine and completely free of moisture/dust. The sky is a deep, dark, saturated blue. The sun is blindingly sharp, and shadows are razor-edged.
- **2.0 to 3.0 (Typical Clear Day):** Standard countryside air. A pleasant, rich blue sky.
- **3.0 to 5.0 (Hazy / Humid / City):** A typical summer day in a city. The horizon turns a washed-out, pale white-blue. The sun develops a noticeable glowing halo.
- **5.0 to 8.0 (Smog / Light Fog):** Heavy industrial pollution or dense morning humidity. The sky shifts toward yellow or grey. Shadows become noticeably soft.
- **8.0 to 10.0+ (Dust Storm / Thick Fog):** Extreme atmospheric scattering. The sky is an opaque brownish-grey. The sun is reduced to a dull, distinct red/orange disc, or vanishes entirely.

Sun Light Intensity

- **High Noon (Clear):** 100,000 to 120,000 Lux
- **Mid-Morning / Afternoon:** 50,000 to 70,000 Lux
- **Overcast / Cloudy Noon:** 10,000 to 20,000 Lux
- **Sunrise / Sunset:** 400 to 1,000 Lux
- **Full Moon:** 0.1 to 0.3 Lux

Sky Light Intensity

- **Clear Blue Sky (Noon):** 15,000 to 20,000 Lux
- **Overcast Sky:** 5,000 to 10,000 Lux
- **Sunrise / Sunset Sky:** 1,000 to 2,000 Lux
- **Clear Night Sky:** 0.001 Lux

Sun Color

- **High Noon (5800K):** #FFFDDFA (Pure white with a microscopic warm tint)
- **Mid-Afternoon (4500K):** #FFF4E6 (Warm golden-white)
- **Golden Hour (3500K):** #FFE0BA (Deep golden orange)
- **Sunset (2500K):** #FFC06D (Fiery orange)

Sky Color

- **Clear Blue Sky (15000K):** #D2E6FF (*Deep, saturated azure*)
- **Overcast / Heavy Clouds (6500K):** #FFFFFF (*Perfectly neutral D65 white*)